

Roll No. :

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: **B.Tech.**

Semester: V

Name of the Course: Introduction to Artificial Intelligence

Course Code: TCS 542

Time : 90 Minutes

Maximum Marks : 50

Note:

- Answer all the questions by choosing any one of the sub questions
- Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	Define Artificial Intelligence. Explain the foundations of AI and their role in building intelligent systems.	
	OR	
(b)	Critically evaluate the current trends in AI (e.g., generative AI, autonomous systems) and their impact on society.	
Q2	(10 Marks)	CO1 & CO3
(a)	Explain the anatomy of an Intelligent System by describing its key components. Also, classify the different types of Intelligent Systems with suitable examples.	
	OR	
(b)	Consider the following graph: <pre> A /\ B C D /\ \ E F G /\ H I</pre> Start Node: A Goal Node: I (a) Perform Depth Limited Search (DLS) on this graph with a depth limit of 2. Show the order of node exploration. (b) Perform Iterative Deepening Search (IDS) for the same graph. Show the order of exploration for each depth until the goal node is found.	
Q3	(10 Marks)	CO2
(a)	Formulate the missionaries and cannibals problem as a state-space problem. Clearly define initial state, goal state, and operators.	
	OR	
(b)	Define state space representation . Give an example of how it can be used to represent the 8-puzzle problem.	
Q4	(10 Marks)	CO3
(a)	An agent is placed in a 2D grid environment where it must find the shortest path from a start state S(0,0) to a goal state G(4,4) . Obstacles are present in certain cells. Explain how the A* search algorithm can be applied to this problem.	

	OR																
(b)	Explain the working principle of the Hill Climbing search algorithm. Discuss its major limitations such as local maxima, plateaus, and ridges.																
Q5	(10 Marks)	CO4															
(a)	(i) Define the terms: random variable, probability distribution and conditional probability. (ii) A coin is tossed twice. Find the probability of getting: (a) Two heads (b) At least one head (c) No heads																
	OR																
(b)	Consider we have two binary variables: Rain (R) = {Yes, No} Sprinkler (S) = {On, Off} (a) What is $P(\text{Rain} = \text{Yes})$? (b) What is $P(\text{Sprinkler} = \text{On})$? (c) What is $P(\text{Rain} = \text{Yes} \mid \text{Sprinkler} = \text{On})$? The joint probability distribution (JPD) is given below <table> <tr> <th>R (Rain)</th><th>S (Sprinkler)</th><th>P(R, S)</th></tr> <tr> <td>Yes</td><td>On</td><td>0.1</td></tr> <tr> <td>Yes</td><td>Off</td><td>0.2</td></tr> <tr> <td>No</td><td>On</td><td>0.3</td></tr> <tr> <td>No</td><td>Off</td><td>0.4</td></tr> </table>	R (Rain)	S (Sprinkler)	P(R, S)	Yes	On	0.1	Yes	Off	0.2	No	On	0.3	No	Off	0.4	
R (Rain)	S (Sprinkler)	P(R, S)															
Yes	On	0.1															
Yes	Off	0.2															
No	On	0.3															
No	Off	0.4															